

CSES Sorting and Searching — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1073 — Towers

Process n cubes in the given order. A cube may start a new tower or be placed on a tower whose current top is strictly larger. Minimize the number of towers. **In:** n ; then $k_1 \dots k_n$

Out: minimum number of towers

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq k_i \leq 10^9$.

1074 — Stick Lengths

Change every stick to one common length. Changing length p_i to x costs $|p_i - x|$. Minimize total cost. **In:** n ; then $p_1 \dots p_n$

Out: minimum total cost

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq p_i \leq 10^9$. Use 64-bit.

1085 — Array Division

Divide an array of positive integers into exactly k contiguous nonempty subarrays. Minimize the largest subarray sum. **In:** n k ; then $x_1 \dots x_n$

Out: minimum possible value of the maximum part sum

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

1091 — Concert Tickets

Tickets have prices h_i . Customers arrive in order with maximum prices t_i . Give each customer the most expensive remaining ticket with price $\leq t_i$; each ticket is sold at most once. **In:** n m ; prices h_i ; customer limits t_i

Out: price paid by each customer, or -1

Constraints: $1 \leq n, m \leq 2 \cdot 10^5$, $1 \leq h_i, t_i \leq 10^9$.

1141 — Playlist

Given a sequence of song IDs, find the longest contiguous segment in which every song ID is distinct. **In:** n ; then $k_1 \dots k_n$

Out: maximum length

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq k_i \leq 10^9$.

1163 — Traffic Lights

A street spans positions 0 through x . Insert n traffic lights one by one at distinct positions. After every insertion, find the longest interval containing no traffic light in its interior. **In:** x n ; then positions $p_1 \dots p_n$

Out: n longest passage lengths, after each insertion

Constraints: $1 \leq x \leq 10^9$, $n \leq 2 \cdot 10^5$, $0 < p_i < x$.

1164 — Room Allocation

Customer i stays from day a_i through b_i . A room can be reused only if the previous customer's departure day is *earlier* than the next arrival day. Minimize rooms and give an allocation. **In:** n ; then a b for each customer

Out: minimum room count k ; then each customer's room in input order

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq a \leq b \leq 10^9$.

1619 — Restaurant Customers

Given arrival and leaving times of n customers, find the maximum number simultaneously in the restaurant. All arrival/leaving times are distinct. **In:** n ; then a b for each customer

Out: maximum simultaneous customers

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq a < b \leq 10^9$.

1620 — Factory Machines

Machine i needs k_i seconds per product and machines work simultaneously. Find the minimum time needed to produce at least t products in total. **In:** n t ; then $k_1 \dots k_n$

Out: minimum required time

Constraints: $n \leq 2 \cdot 10^5$, $t \leq 10^9$, $k_i \leq 10^9$. Use 64-bit.

1629 — Movie Festival

Given start/end times of n movies, choose a maximum-size set that one person can watch completely without overlaps. **In:** n ; then a b for each movie

Out: maximum number of movies

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq a < b \leq 10^9$.

1630 — Tasks and Deadlines

Process every task exactly once, sequentially from time 0. Task i has duration a_i and deadline d_i ; if it finishes at f_i , its reward is $d_i - f_i$. Maximize total reward. **In:** n ; then a d for each task

Out: maximum total reward

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a, d \leq 10^6$. Use 64-bit.

1631 — Reading Books

Two readers must each read all n books from beginning to end. Book i takes t_i time, and the two readers cannot read the same book simultaneously. Minimize the total finishing time. **In:** n ; then $t_1 \dots t_n$

Out: minimum total time

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq t_i \leq 10^9$. Use 64-bit.

1632 — Movie Festival II

There are k club members and n movies with start/end times. Each member may watch only non-overlapping movies. Maximize the total number watched across all members. **In:** n k ; then a b for each movie

Out: maximum total number of movies

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq a < b \leq 10^9$.

1640 — Sum of Two Values

Find two array values at distinct positions whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: two 1-indexed positions, or IMPOSSIBLE

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x, a_i \leq 10^9$.

1641 — Sum of Three Values

Find three array values at distinct positions whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: three 1-indexed positions, or IMPOSSIBLE

Constraints: $1 \leq n \leq 5000$, $1 \leq x, a_i \leq 10^9$.

1642 — Sum of Four Values

Find four array values at distinct positions whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: four 1-indexed positions, or IMPOSSIBLE

Constraints: $1 \leq n \leq 1000$, $1 \leq x, a_i \leq 10^9$.

1643 — Maximum Subarray Sum

Find the maximum sum of a contiguous, nonempty subarray. **In:** n ; then $x_1 \dots x_n$

Out: maximum subarray sum

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $-10^9 \leq x_i \leq 10^9$. Use 64-bit.

1644 — Maximum Subarray Sum II

Find the maximum sum of a contiguous subarray whose length is between a and b , inclusive. **In:** n a b ; then $x_1 \dots x_n$

Out: maximum valid subarray sum

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a \leq b \leq n$, $|x_i| \leq 10^9$. Use 64-bit.

1645 — Nearest Smaller Values

For every array position i , find the nearest position $j < i$ with $x_j < x_i$. **In:** n ; then $x_1 \dots x_n$

Out: n positions; print 0 when no such j exists

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

1660 — Subarray Sums I

Given an array of *positive* integers, count contiguous subarrays whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: number of matching subarrays

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x, a_i \leq 10^9$. Use 64-bit count.

1661 — Subarray Sums II

Given an integer array, count contiguous subarrays whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: number of matching subarrays

Constraints: $n \leq 2 \cdot 10^5$, $-10^9 \leq x, a_i \leq 10^9$. Use 64-bit.

1662 — Subarray Divisibility

Count contiguous subarrays whose sum is divisible by n . **In:** n ; then $a_1 \dots a_n$

Out: number of valid subarrays

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $-10^9 \leq a_i \leq 10^9$. Use 64-bit.

2162 — Josephus Problem I

Children $1, 2, \dots, n$ stand in a circle. Repeatedly remove every other remaining child until none remain.

In: n

Out: the complete removal order

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

2163 — Josephus Problem II

Children $1, 2, \dots, n$ stand in a circle. Repeatedly skip k remaining children and remove the next child. **In:** n k

Out: the complete removal order

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $0 \leq k \leq 10^9$.

2168 — Nested Ranges Check

For each distinct range $[x, y]$, determine (1) whether it contains some other input range, and (2) whether some other input range contains it. $[a, b]$ contains $[c, d]$ iff $a \leq c$ and $d \leq b$. **In:** n ; then x y for each range

Out: two n -entry 0/1 lines, both in input order: contains; contained-by

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x < y \leq 10^9$; ranges distinct.

2169 — Nested Ranges Count

For each distinct range $[x, y]$, count (1) how many other ranges it contains and (2) how many other ranges contain it, using inclusive containment. **In:** n ; then x y for each range

Out: two n -entry lines in input order: contained count; containing count

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x < y \leq 10^9$; ranges distinct.

2183 — Missing Coin Sum

Given positive coin values, each usable at most once, find the smallest positive integer sum that cannot be formed by a subset. **In:** n ; then $x_1 \dots x_n$

Out: smallest unformable positive sum

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

2216 — Collecting Numbers

The array is a permutation of $1, \dots, n$. In each round scan left-to-right, collecting as many next required numbers as possible while collecting $1, 2, \dots, n$ in order. Find the number of rounds. **In:** n ; then permutation $x_1 \dots x_n$

Out: number of rounds

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

2217 — Collecting Numbers II

Same collecting process as above. Perform m operations, each swapping the values at two positions a, b . Report the required number of rounds after every swap.

In: n m ; permutation; then m swaps a b

Out: m answers, one after each swap

Constraints: $1 \leq n, m \leq 2 \cdot 10^5$, $1 \leq a, b \leq n$.

2428 — Distinct Values Subarrays II

Count contiguous subarrays containing at most k distinct values. **In:** n k ; then $x_1 \dots x_n$

Out: number of valid subarrays

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3420 — Distinct Values Subarrays

Count contiguous subarrays in which every element is distinct. **In:** n ; then $x_1 \dots x_n$

Out: number of all-distinct subarrays

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3421 — Distinct Values Subsequences

Count nonempty subsequences in which every chosen element is distinct. Subsequences are determined by positions and may have gaps. Print the count modulo $10^9 + 7$. **In:** n ; then $x_1 \dots x_n$

Out: number of all-distinct subsequences mod $10^9 + 7$

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.
